

WEEK 2 Outline

Planetary

Astronomy

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The Naked – Eye Universe

Constellations and Asterisms

Constellation: a designated region in the sky containing one or more historical star patterns.

Examples) Orion, Taurus, Ursa Major (see next slides)

Asterism: a recognizable pattern of stars.

Ex) Orion, the hunter

Ex) Taurus, the bull; the Pleiades; the Hyades

Ex) Ursa Major (the great bear); the Big Dipper; La Cassarole

Ex) The Summer Triangle

Ex) The Coathanger (Brocchi's cluster)

- * 88 total constellations

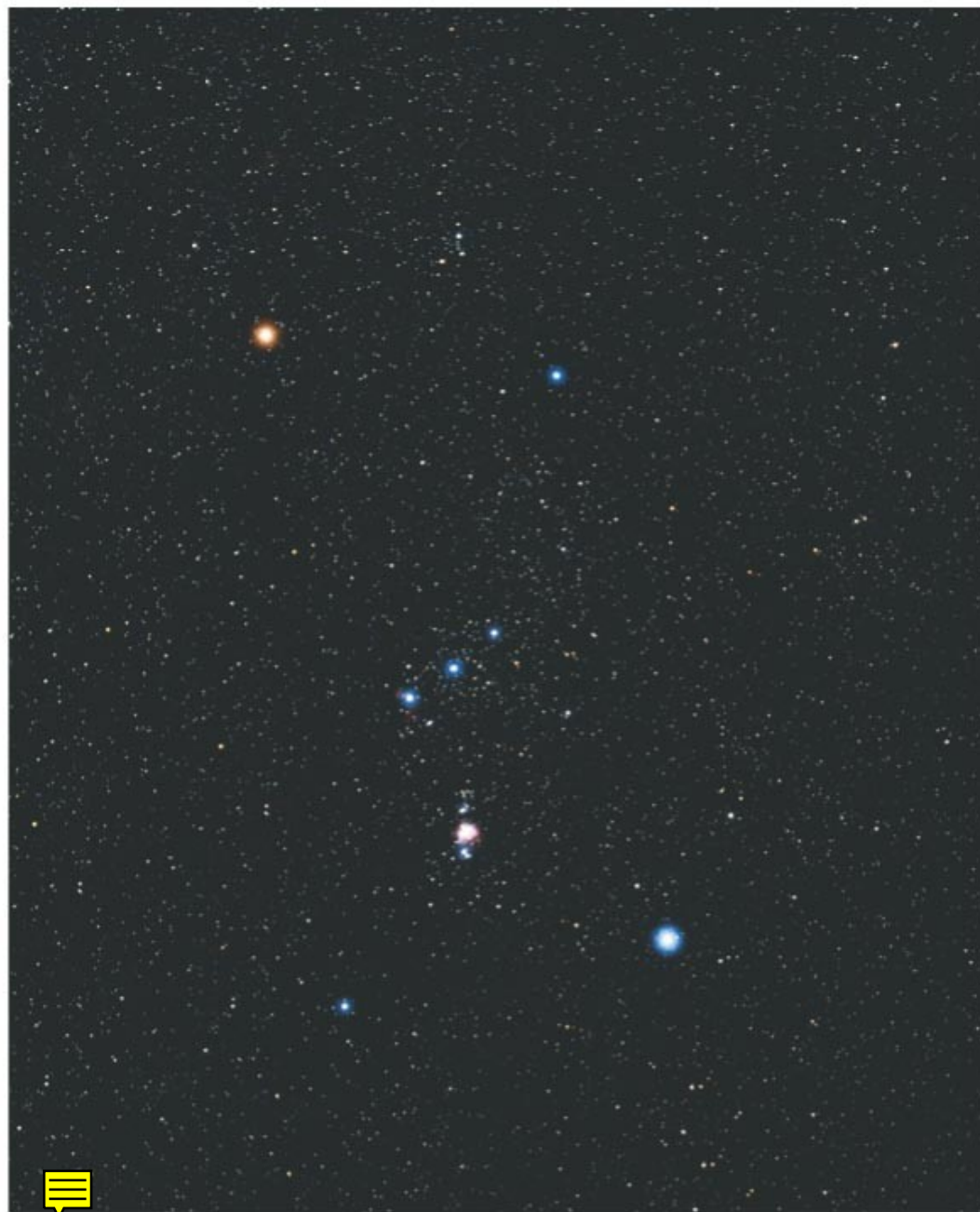
- * More than 88 asterisms

- * Northern constellations named after Greek Mythological characters

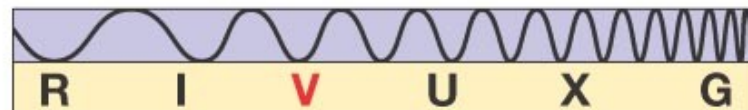
**Example:
Orion.**

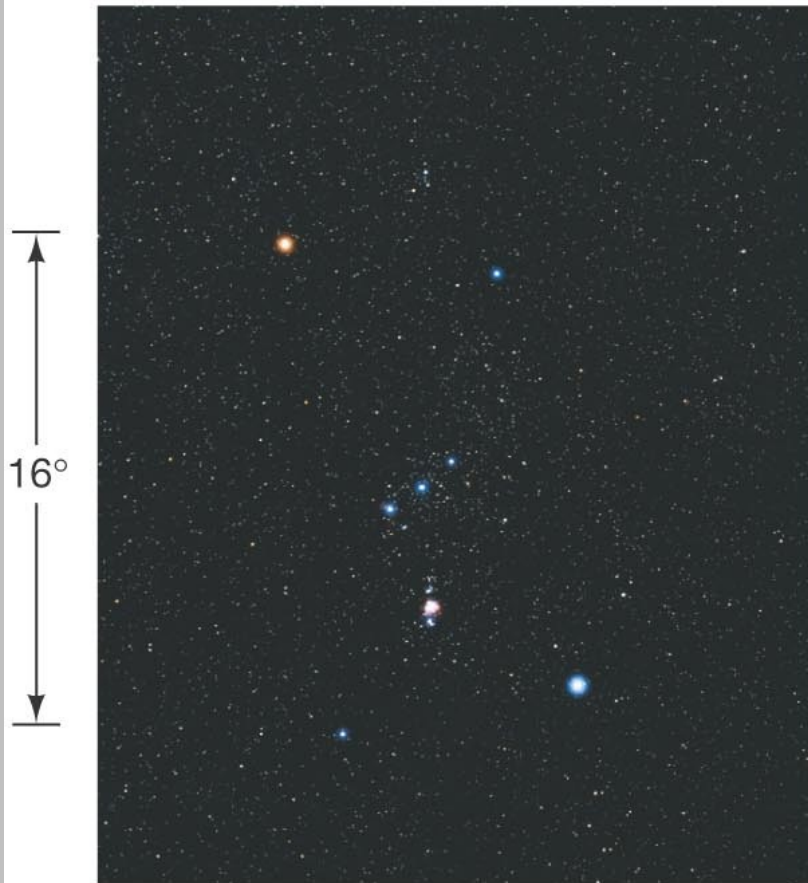
**An easily
recognized
constellation!**

16°

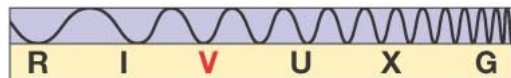


(a)

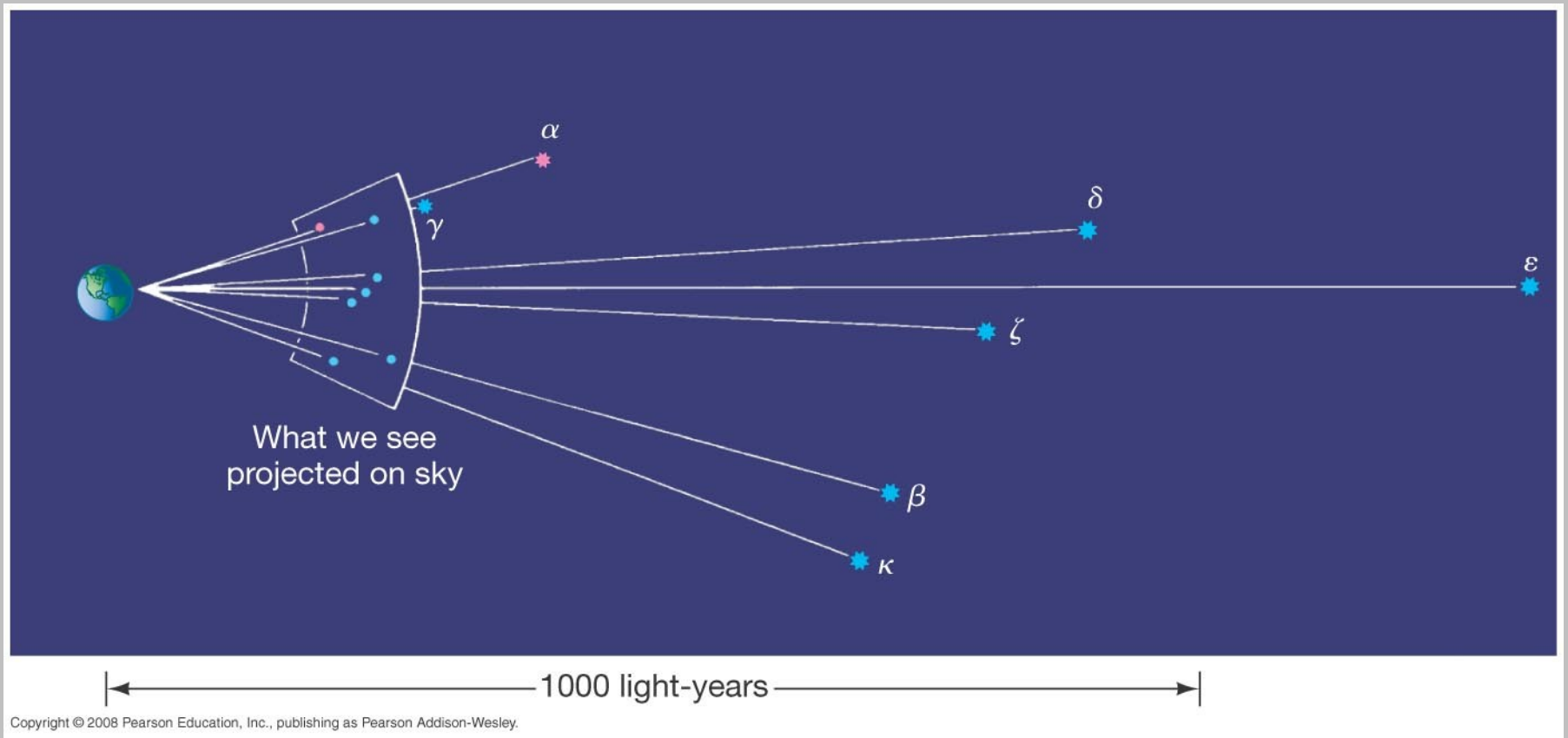




(a)



(b)



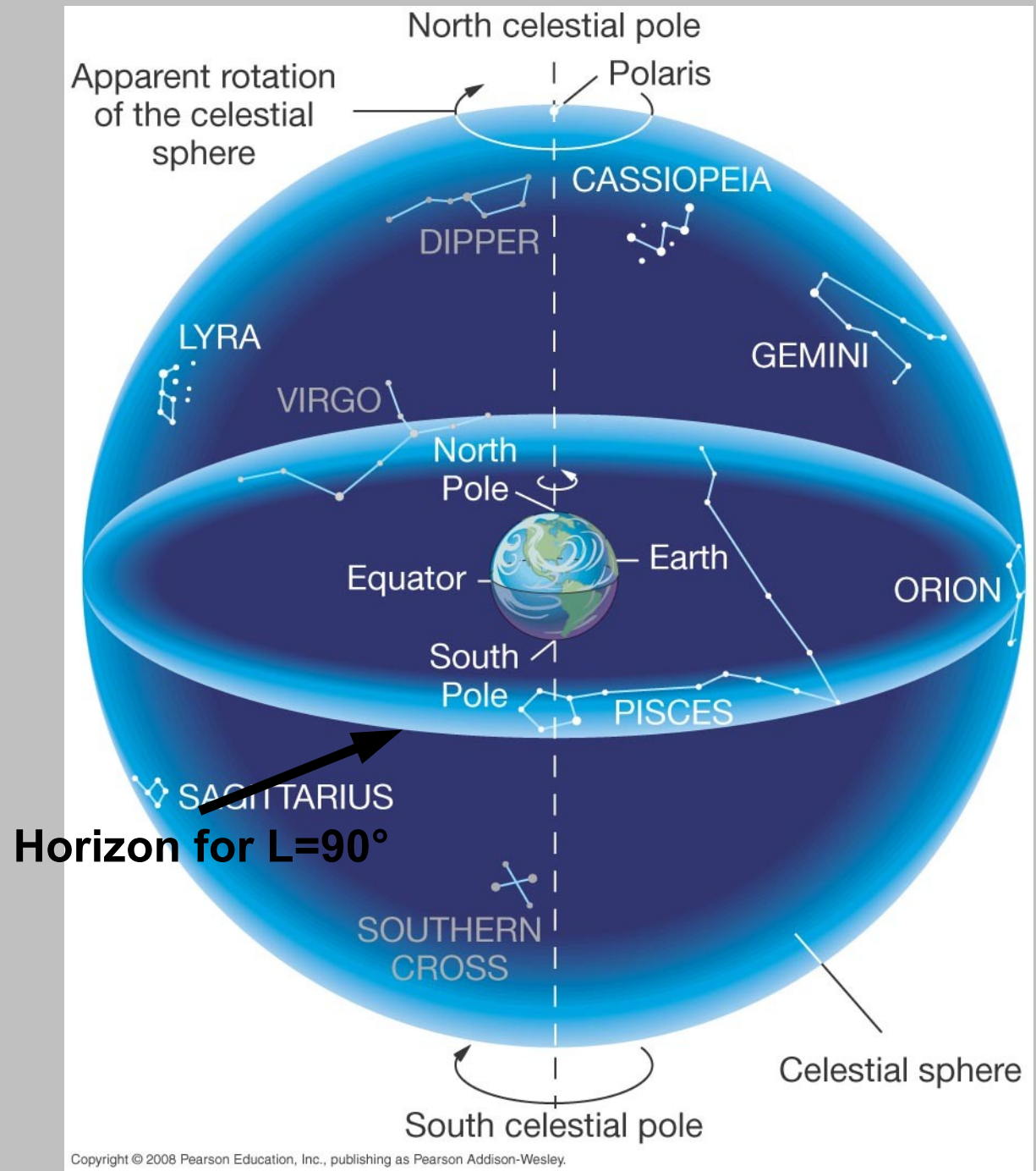


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Try a planetarium program like “Stellarium” to see the sky in motion.

The Celestial Sphere

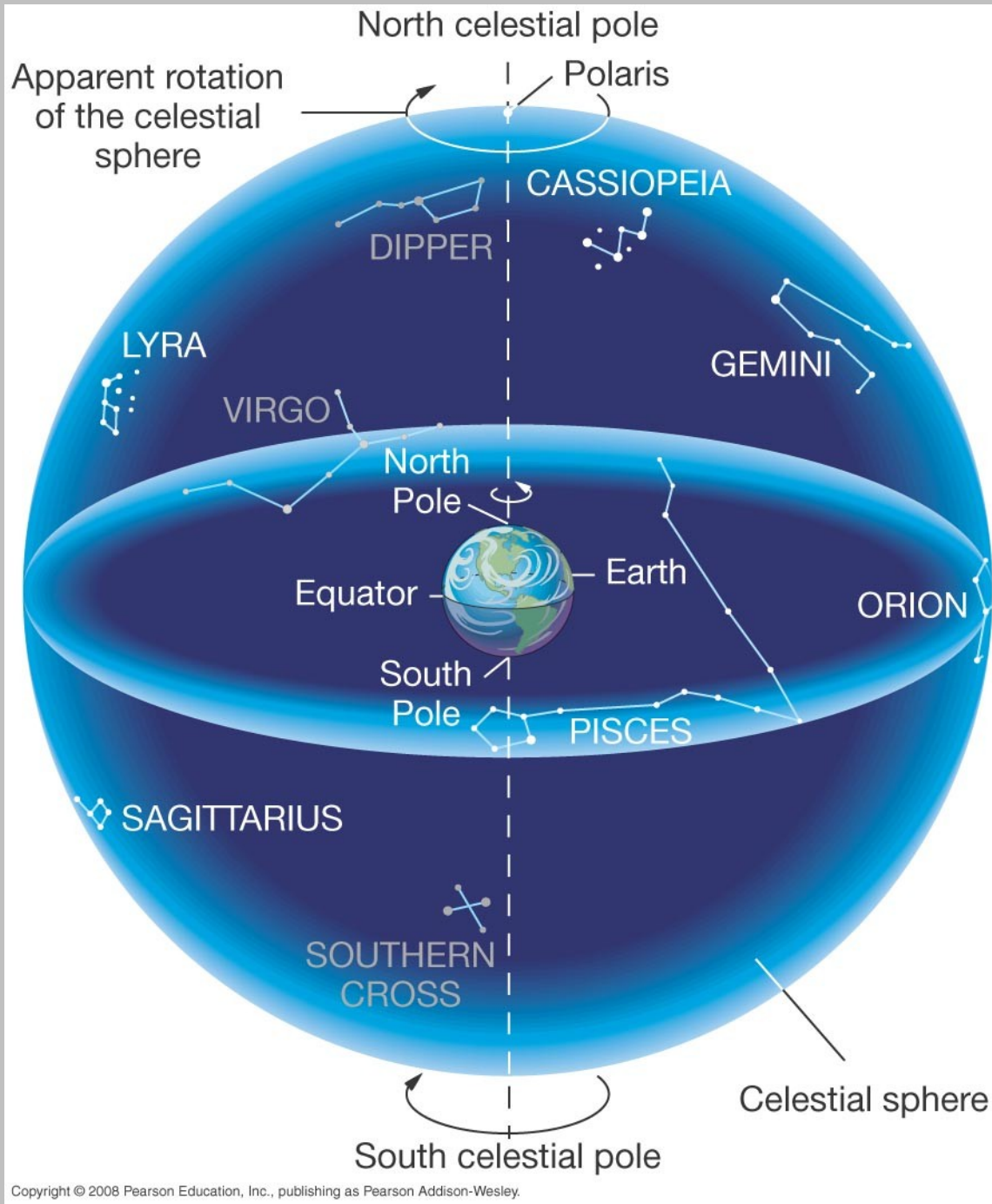
- a conceptual model of the sky.
- geocentric (wrong)
- all stars at same distance (wrong)
- a distortion-free sky map
- reproduces daily rising and setting motions for any latitude on Earth
- Cel. Sphere is infinitely bigger than the Earth.



The Celestial Sphere

Features:

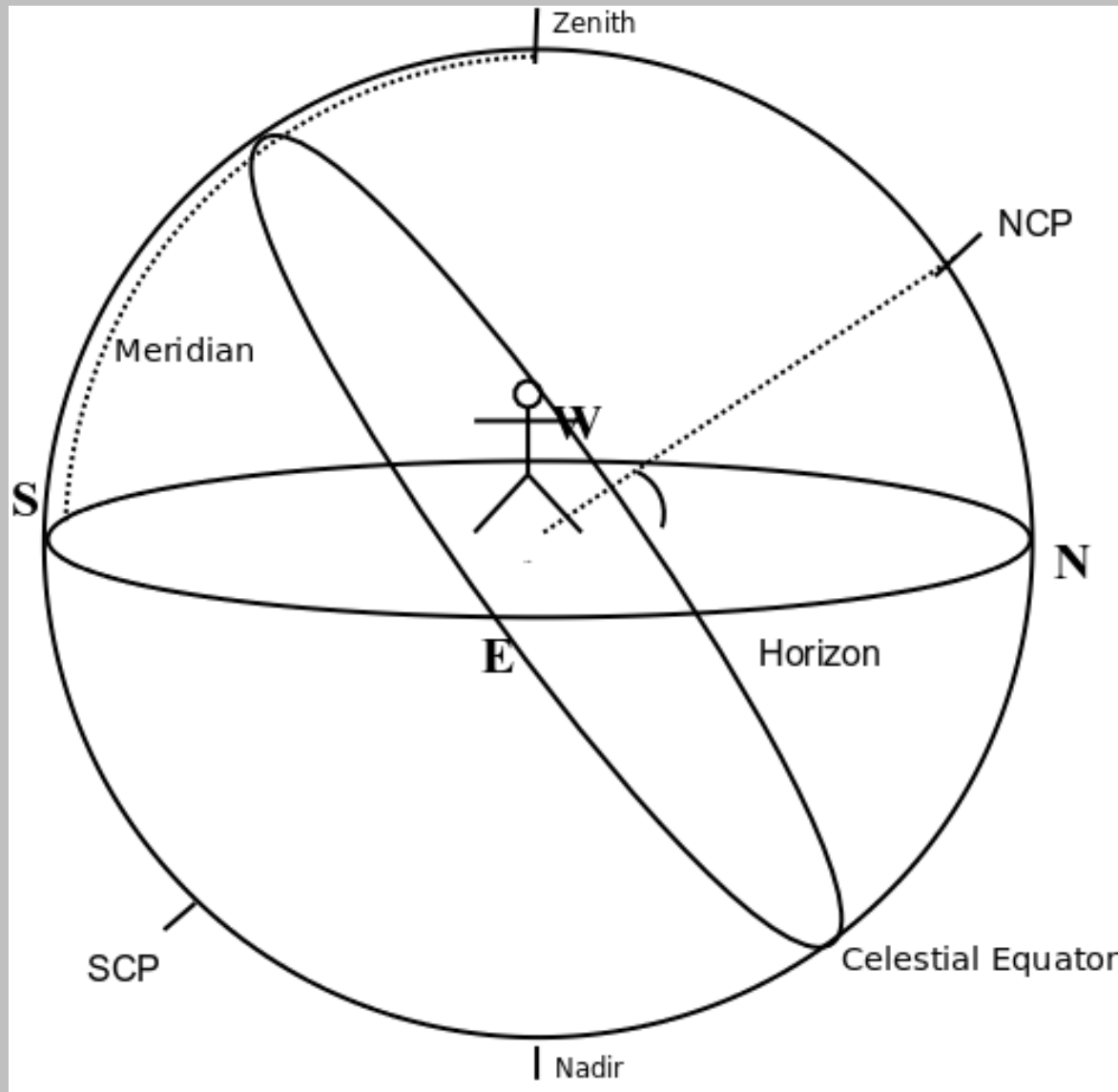
1. stars
2. Earth/observer
3. N. Celestial Pole
4. S. Celestial Pole
5. Celestial Equator



The Celestial Sphere

More Features:

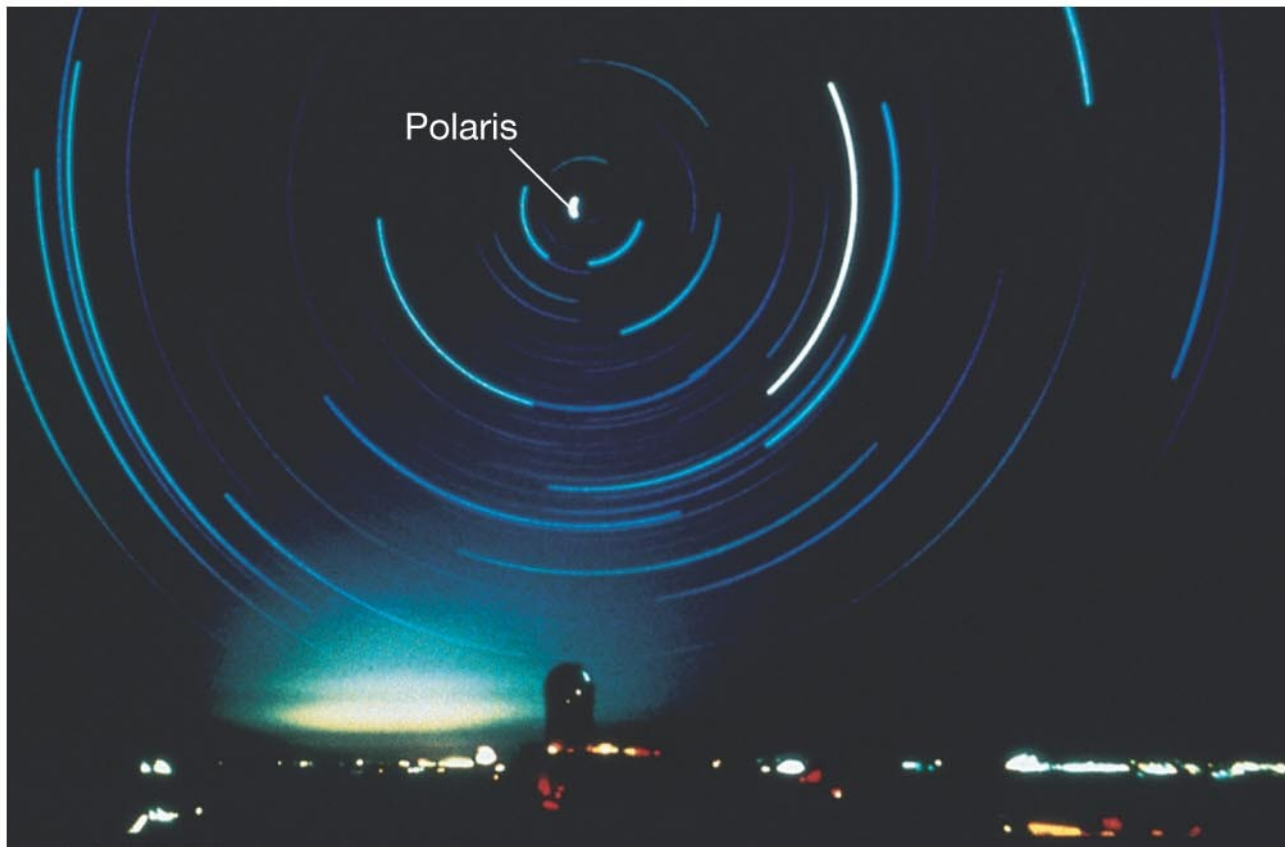
1. stars
2. Earth/observer
3. N. Celestial Pole
4. S. Celestial Pole
5. Celestial Equator
6. Horizon
7. Cardinal points, (N,S,E,W)
8. Zenith
9. Nadir
10. Meridian



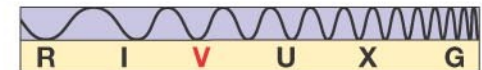
See astro.unl.edu 's [horizon diagram switch](#).

Motion of the Earth - Daily

Star Trail – an actual photo of Northern horizon. Exposure time was about 5 hours.



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Now show the daily rotation with a Celestial Globe.

Motion of the Earth – Daily

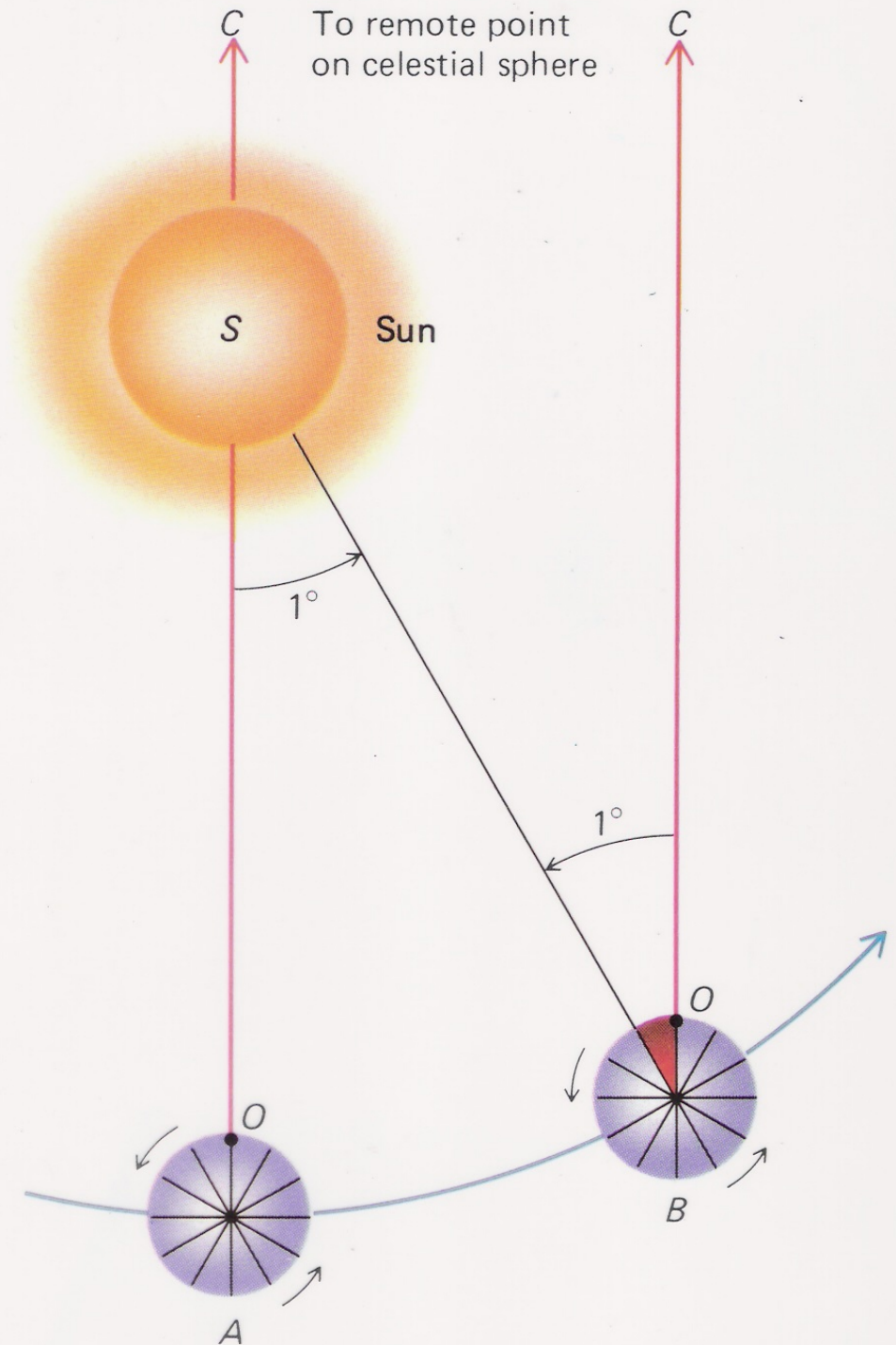
The Day

Sidereal Day: the time that it takes for the Earth to rotate 360 degrees relative to the distant stars

Mean Solar Day: the average time that it takes for the Earth to rotate relative to the Sun. (E.g., from one noon to the next.)

- * What your watch tells you
- * 3 m 56 s longer than the sidereal day

- * “Mean” because the time between local noons varies.
 - Obliquity (tilt) of ecliptic
 - Elliptical orbit



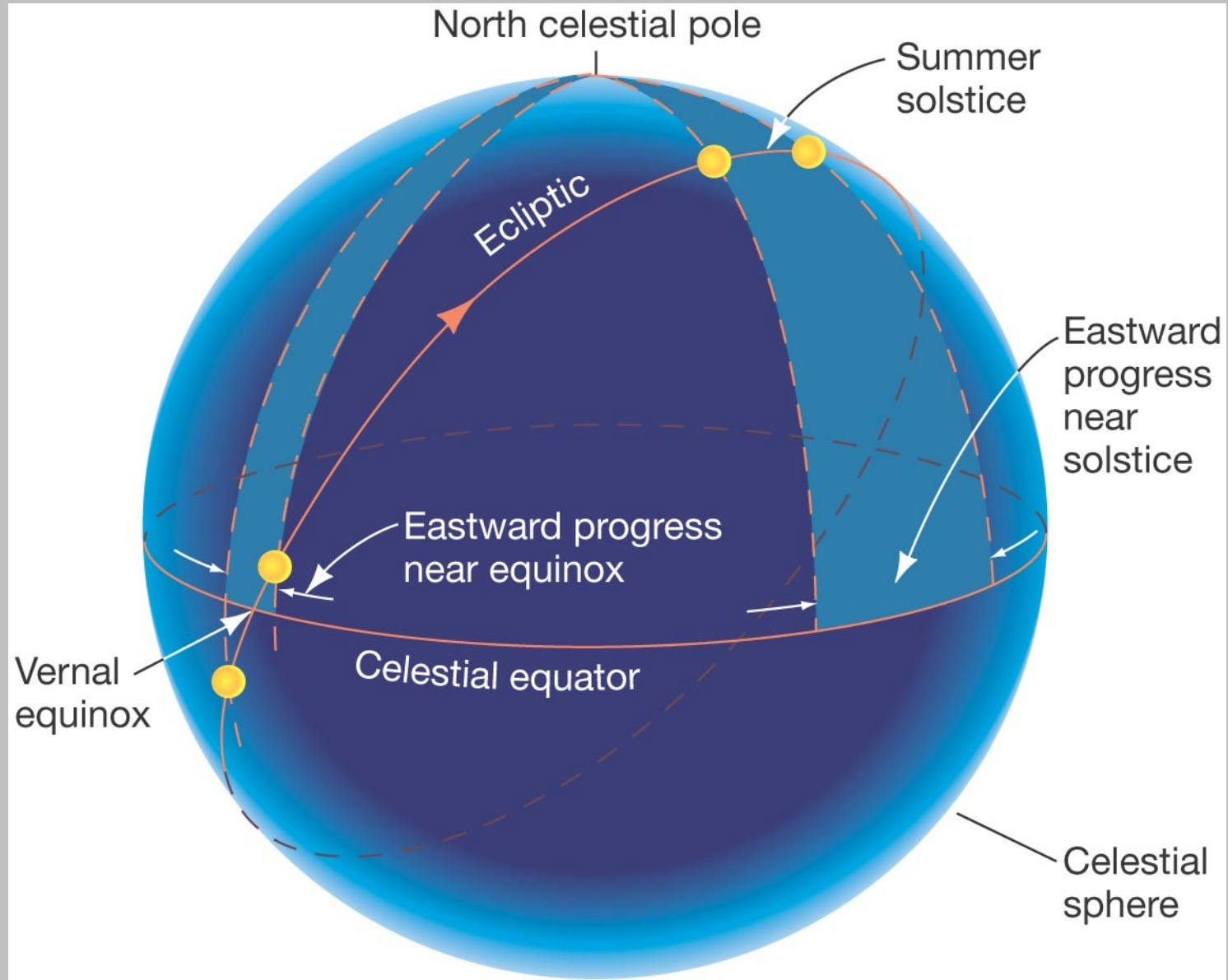
Motion of the Earth - Daily

Since the ecliptic is tilted, the rate of the Sun's apparent motion

Eastward
relative to the
stars changes.

(It is fast during the solstices and slow during the equinoxes.)

**So the length of
the Apparent
Solar Day
changes.**



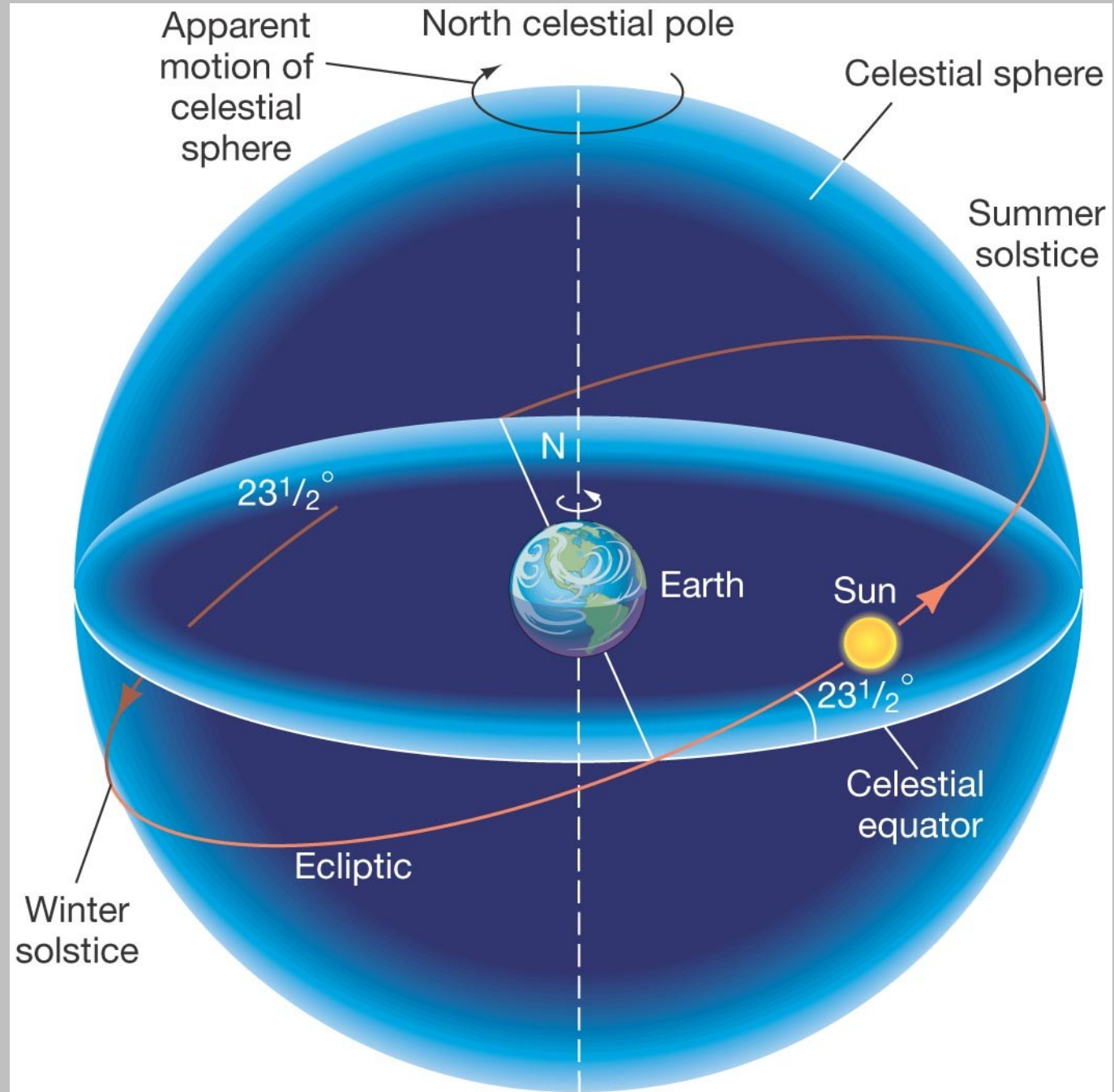
(b)

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The ASD is *longer* during solstices

Motion of the Earth - Annual

The Sun appears to go around the Earth once per year, but it is the Earth that goes around the Sun.



Motion of the Earth – Annual

The Year

Sidereal year: the time it takes for the Earth to revolve around the Sun with respect to the stars.

* 365.2564 mean solar days

Tropical Year = time between two successive passages of the Sun past the *Vernal Equinox*

* 365.2422 days

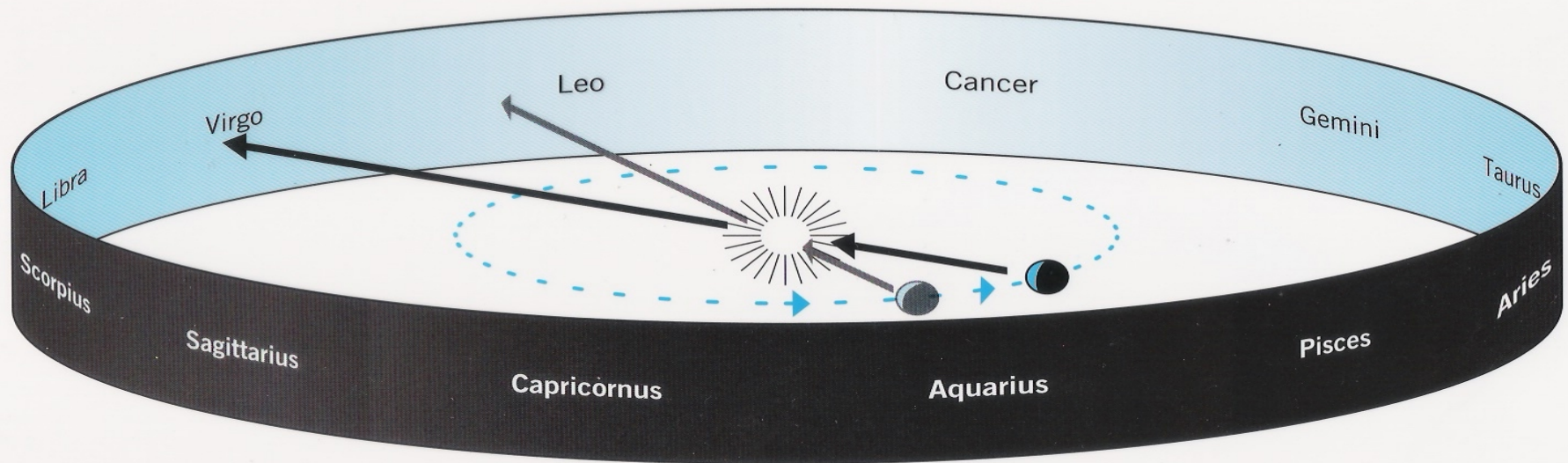
Q: Why is the tropical year shorter than the sidereal year?

A: Precession

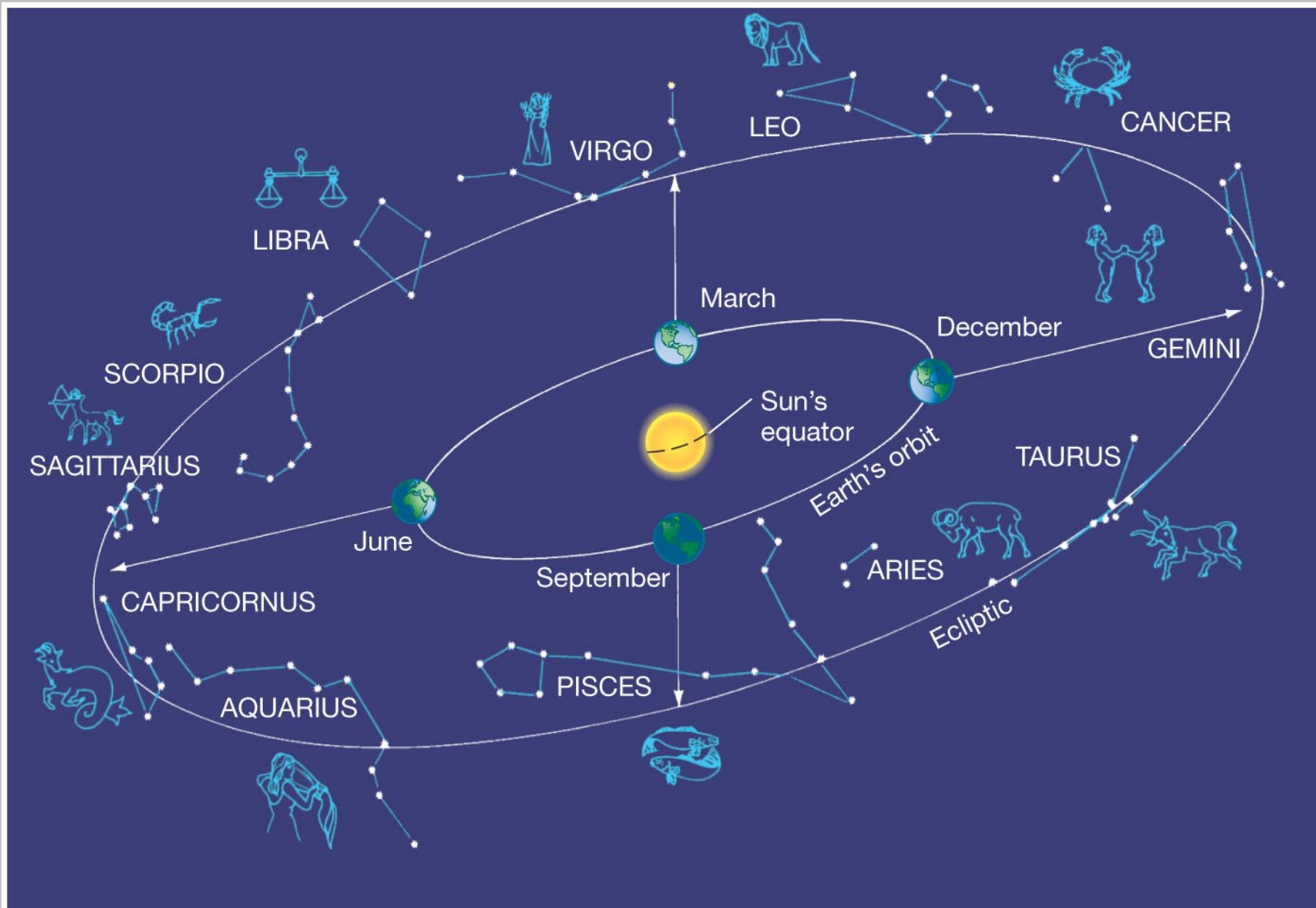
Motion of the Earth – Annual

The Zodiac = the 12 (or 13 counting Ophiuchus) constellations through which the Sun passes in a year.

Ecliptic = The apparent path of the Sun on the sky as seen from Earth.



Motion of the Earth - Annual

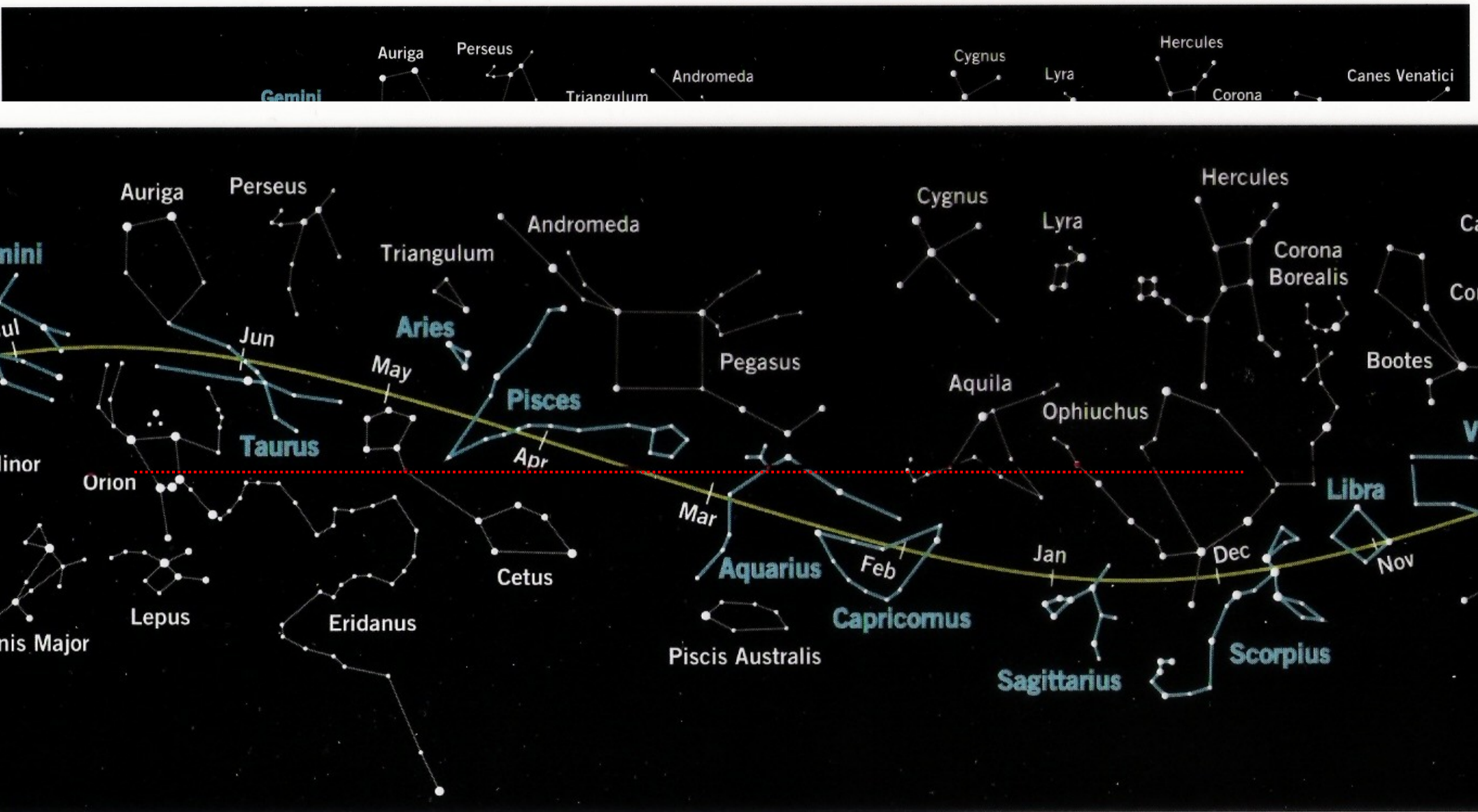


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The Zodiacal Constellations

Motion of the Earth – Annual

Ecliptic = The apparent path of the Sun on the sky as seen from Earth.



Motion of the Earth and Seasons

Seasons: an oscillation of average temperature with a period of 1 tropical year.

There are two main reasons for the seasons:

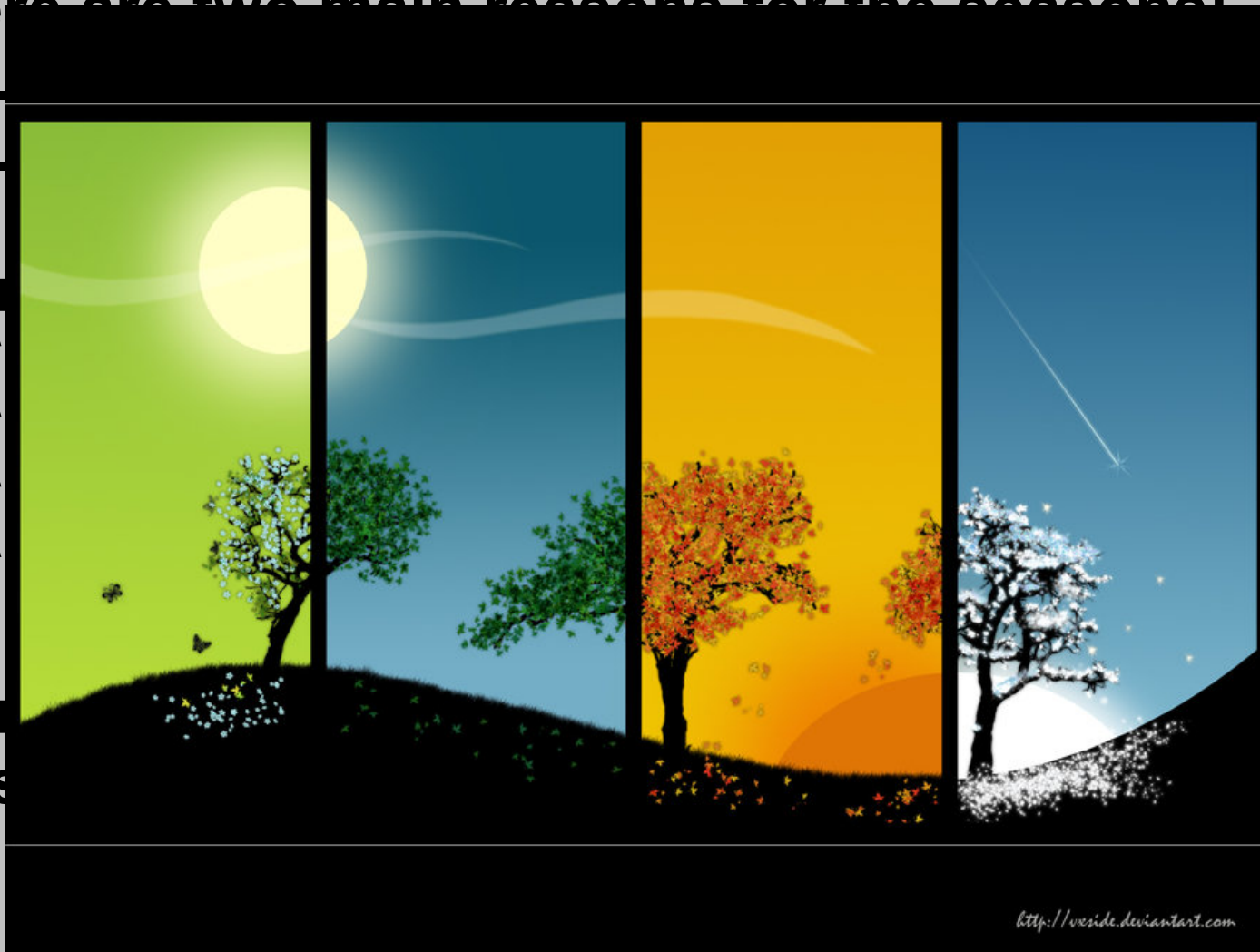
1. **Earth's axial tilt causes changes.**

2.

Astr

*
*
*
*

**Equ
Sols**

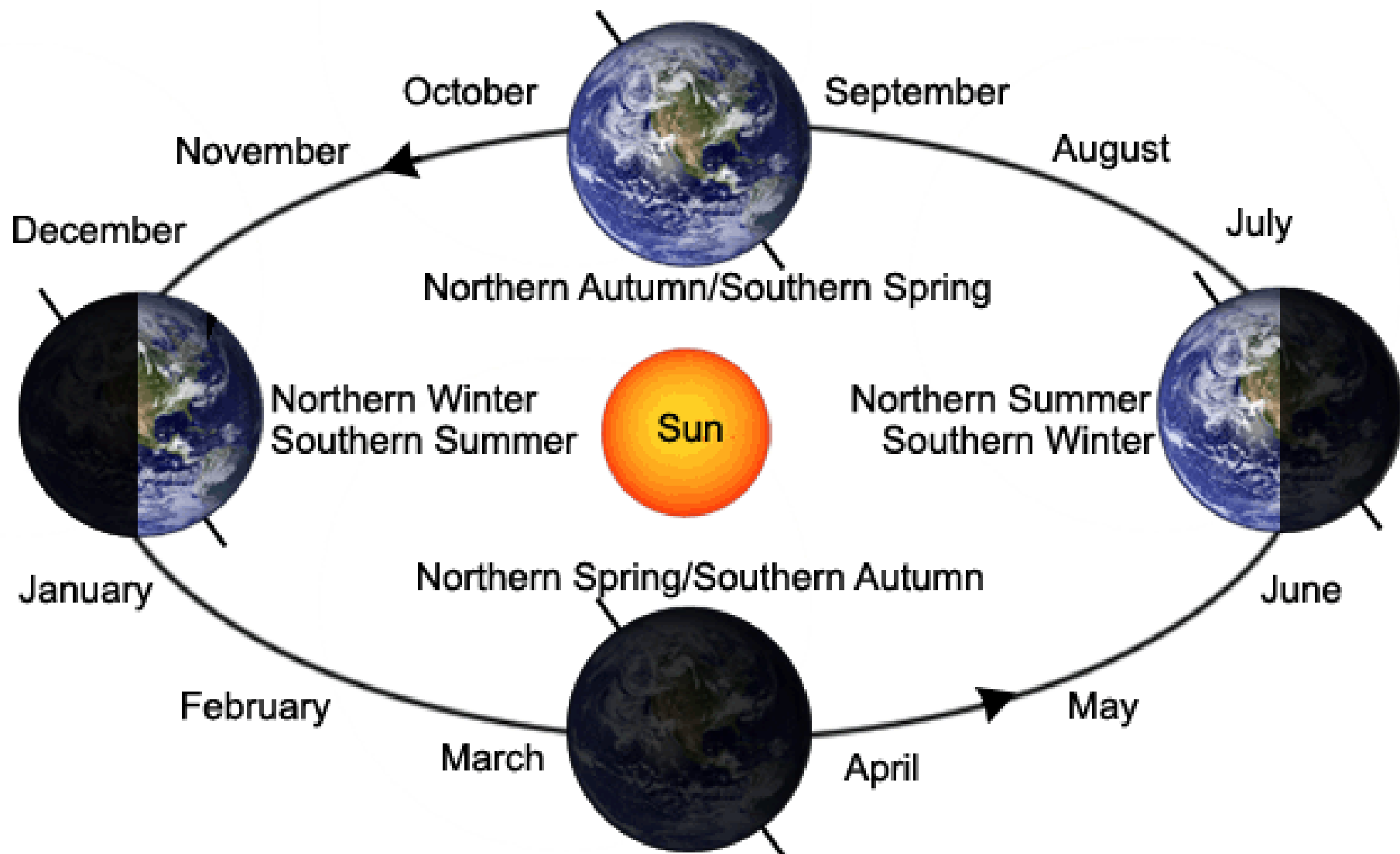


21)
e 20,21)
2, 23)
22)

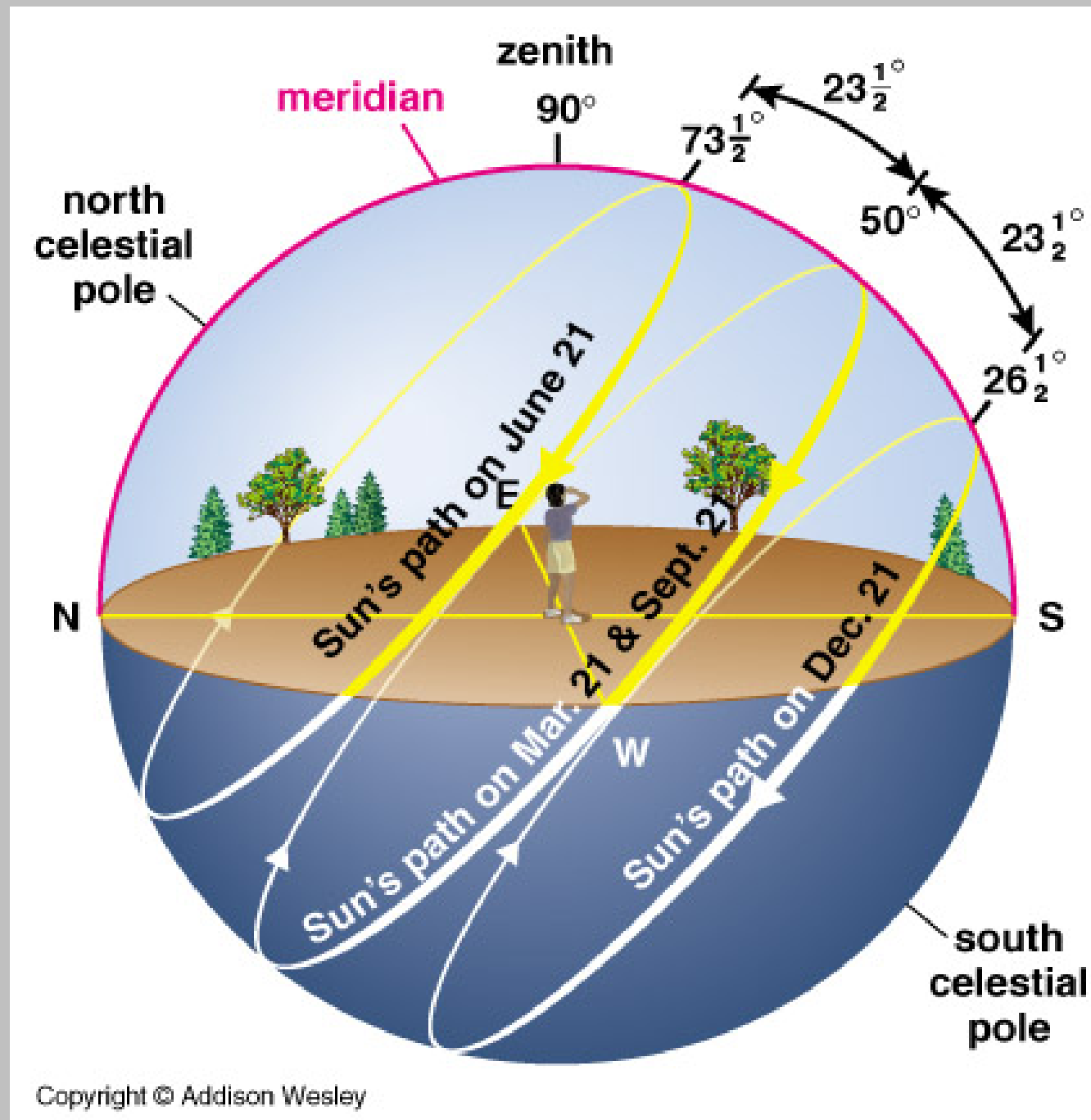
uator!
C.E.!

Motion of the Earth and Seasons

Factor 1: directness of Sun's rays.



Factor 2: length of daylight hours.



Cause of seasons (cont.).

The astro.unl.edu site offers several useful simulators to help you understand both Factor's 1 and 2. (You must unblock Flash.)

Factor 1 (directness of Sun's rays):

Sun's rays simulator

Seasons and ecliptic simulator

Factor 2 (length of daylight hours):

Daylight Simulator

Both:

Sun Motions

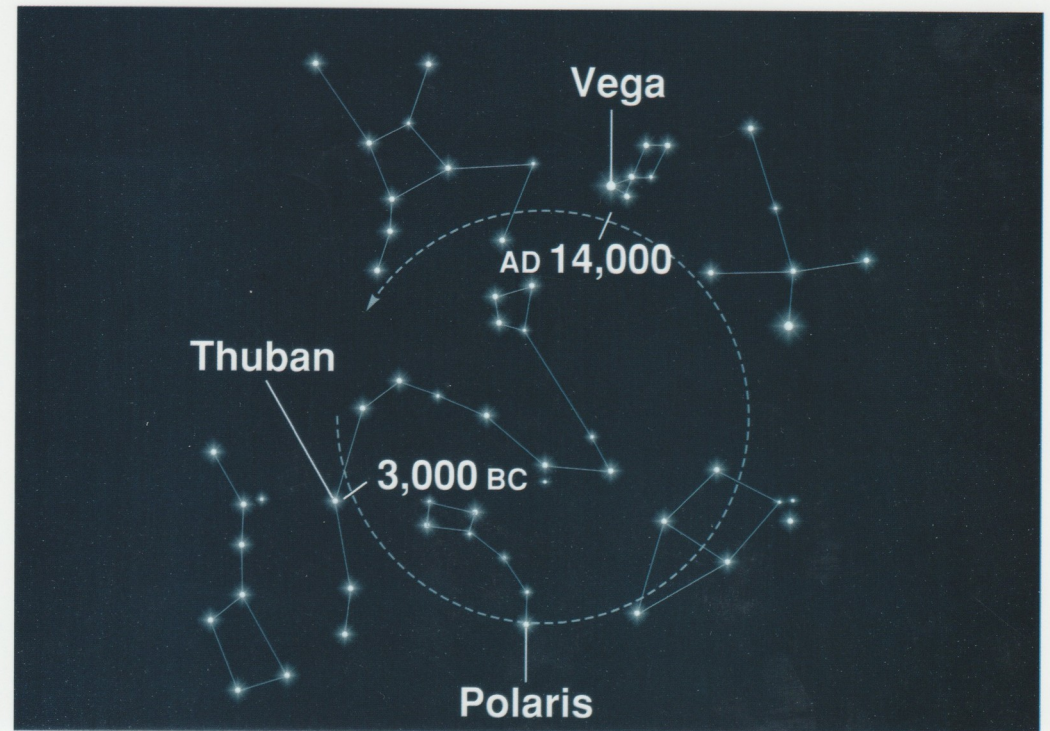
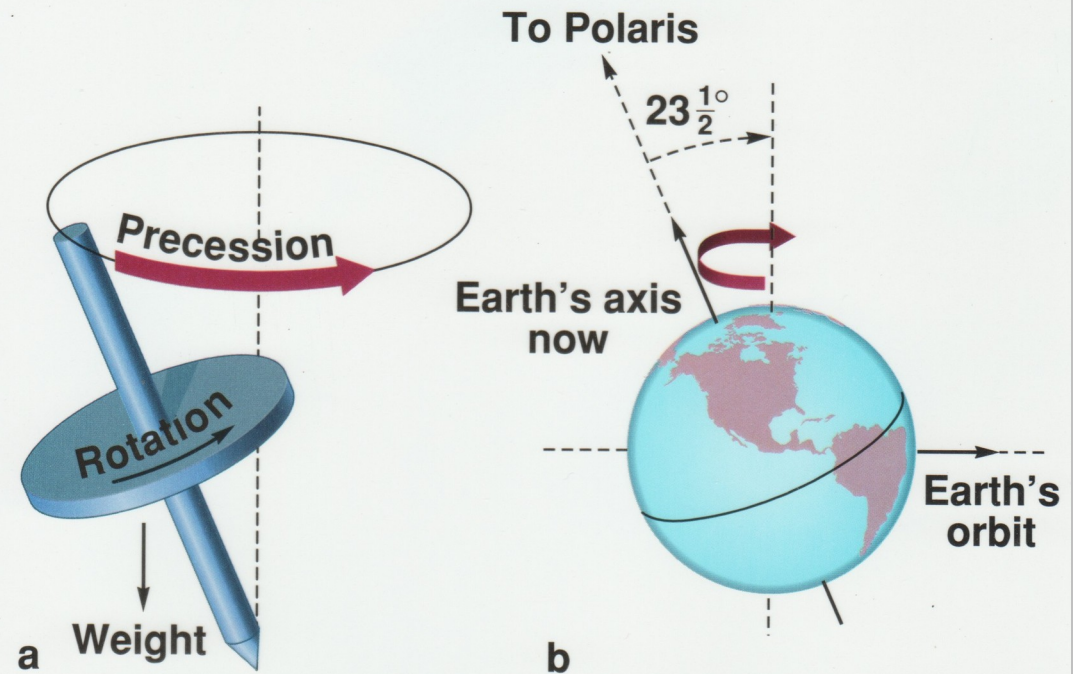
Precession of the Equinoxes – *the reason the calendar year is different than the sidereal year.*

The Earth's spin axis is “wobbling”.

- **26,000 year period**
- **Tilt remains 23.5 degrees**
- **Seasons remain the same severity**

Precession

Gradual change of the
NCP position.
Polestar was not always
Polaris!

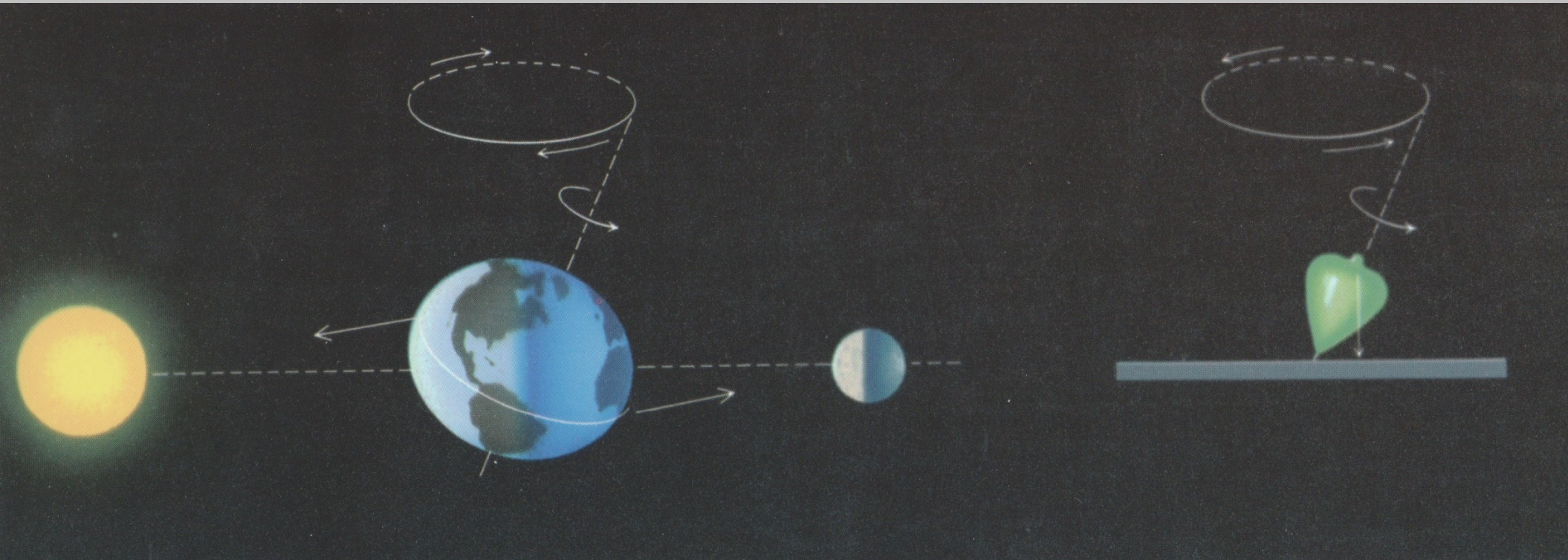


Precession



Precession

Cause: the pull of the Moon and Sun on Earth's equatorial bulge exerts a Torque.



**Left: gravity from S and M are trying to tip the spin axis UPRIGHT.
Pole precesses CW seen from above.**

**Right: gravity is trying to tip the spin axis OVER.
Pole precesses CCW seen from above.**

Precession (of the Equinoxes)

Consequences

- 1) The NCP is changing position rel to stars**
- 2) The equatorial coordinates (RA and DEC) of stars slowly change with time. → We need to specify “Epoch” of coordinates.**

Coordinate Systems for the sky

Altazimuth coordinate system

Uses the horizon for it's zeropoints.

A star's coordinates are different for observers on different parts of Earth

Altitude = angle measured above (or below) the horizon in degrees.

Azimuth = angle measured along the horizon in degrees such that 0° azimuth is due North, 90° is due East, etc.

Ex) Polaris

Altitude = 40.75 degrees (our latitude)

Azimuth = 0 degrees (straight above N on horizon)

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Equatorial (or Celestial) coordinate system

Uses the Celestial equator and ecliptic to define zeropoints.

A star's coordinates are the same for all observers!

Right Ascension, RA = distance measured in hours, minutes and seconds along the celestial equator such that RA=0h at the vernal equinox and RA=6h at the Summer Solstice.

Declination, DEC = Angle measured in degrees, arcminutes and arcseconds above the celestial equator such that DEC = 0° on the cel. equator, increasing to $+90^\circ$ at the NCP and -90° at the SCP.

Coordinate Systems for the sky

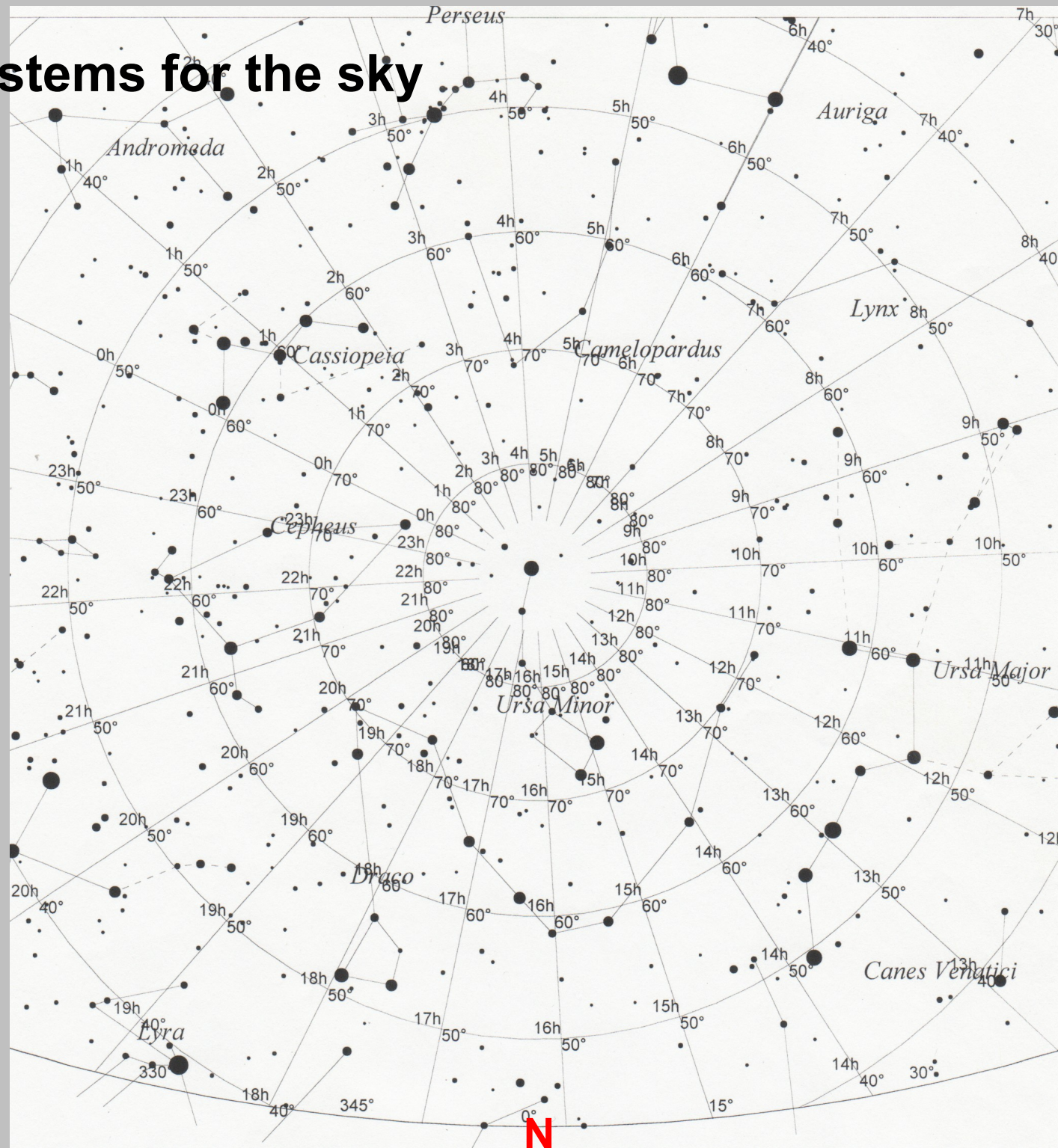
Example 1)

Equatorial:

Polaris at
Dec = 90°
RA = $\sim 2\text{hr}$

Altazimuth:

Polaris has
Alt = 46°
Azim = 0°



Coordinate Systems for the sky

Example 2)

Equatorial:

Saturn has

Dec = -1°

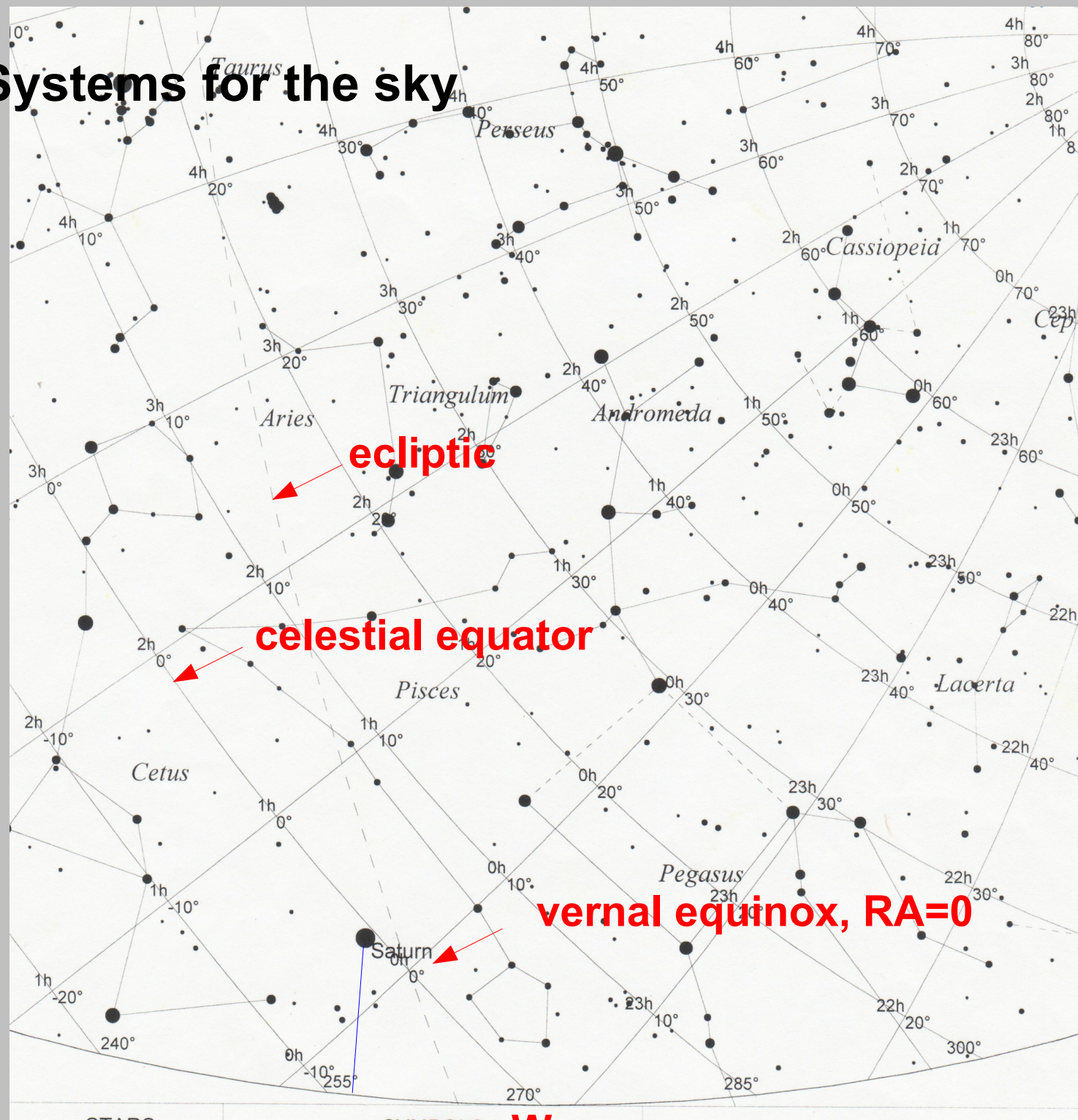
RA = 0h 15m

Altazimuth:

Saturn has

Alt = 13°

Azim = 257°



W

Angles, distances and widths

Angles are measured in degrees, arcminutes, and arcseconds.

1 degree ($^{\circ}$) is $1/360$ of a complete rotation

1 arcminute ($'$) is $1/60$ of a degree

1 arcsecond ($''$) is $1/60$ of an arcminute

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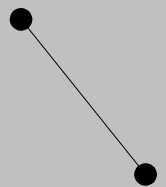
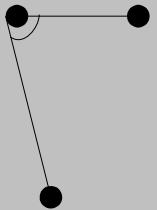
1 arcsecond ($''$) is $1/60$ of an arcminute

Angles on the sky can be measured in two ways:

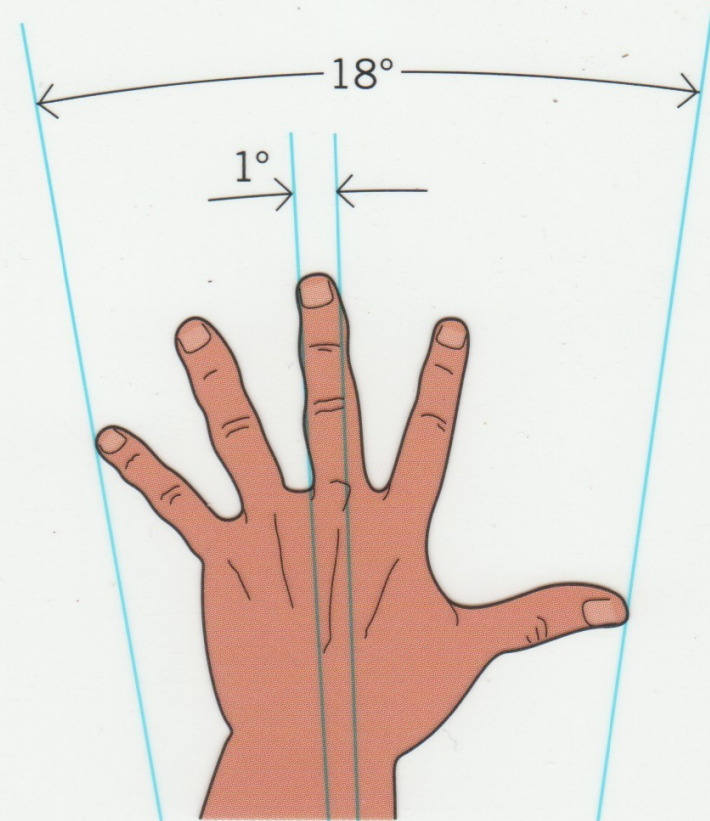
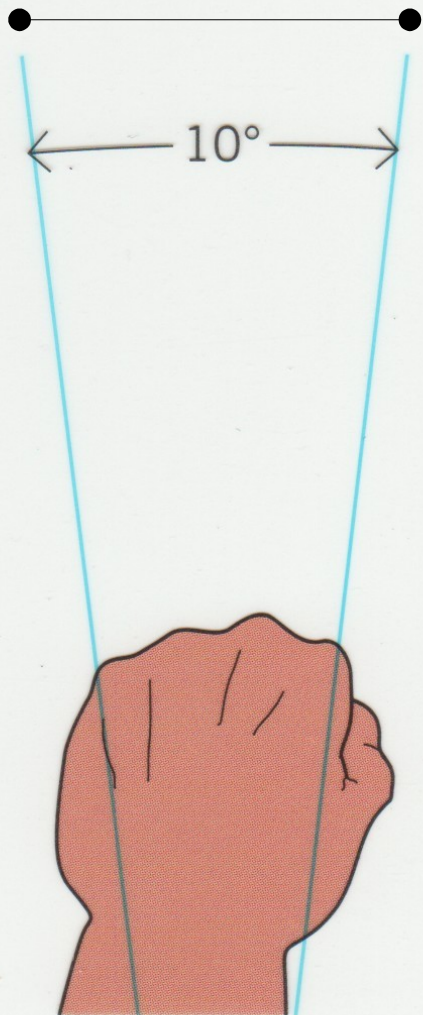
1) Think of the sky as a flat paper with lines connecting one star to two others. An angle can be drawn between those two lines where they intersect.

2) angular separations between two stars as you would estimate using your fist at arm's length.

Mostly we use #2.

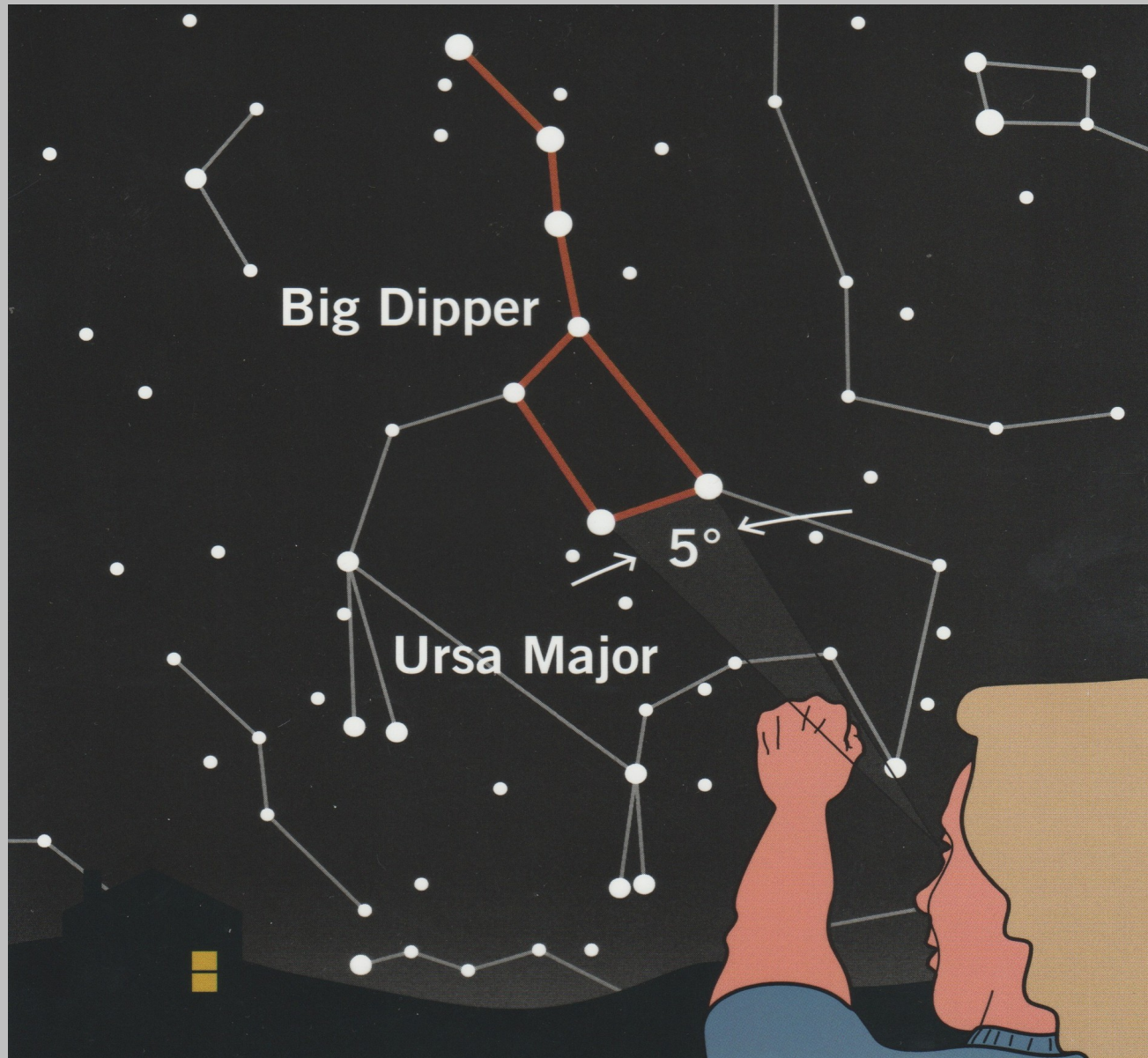


Angles, distances and widths



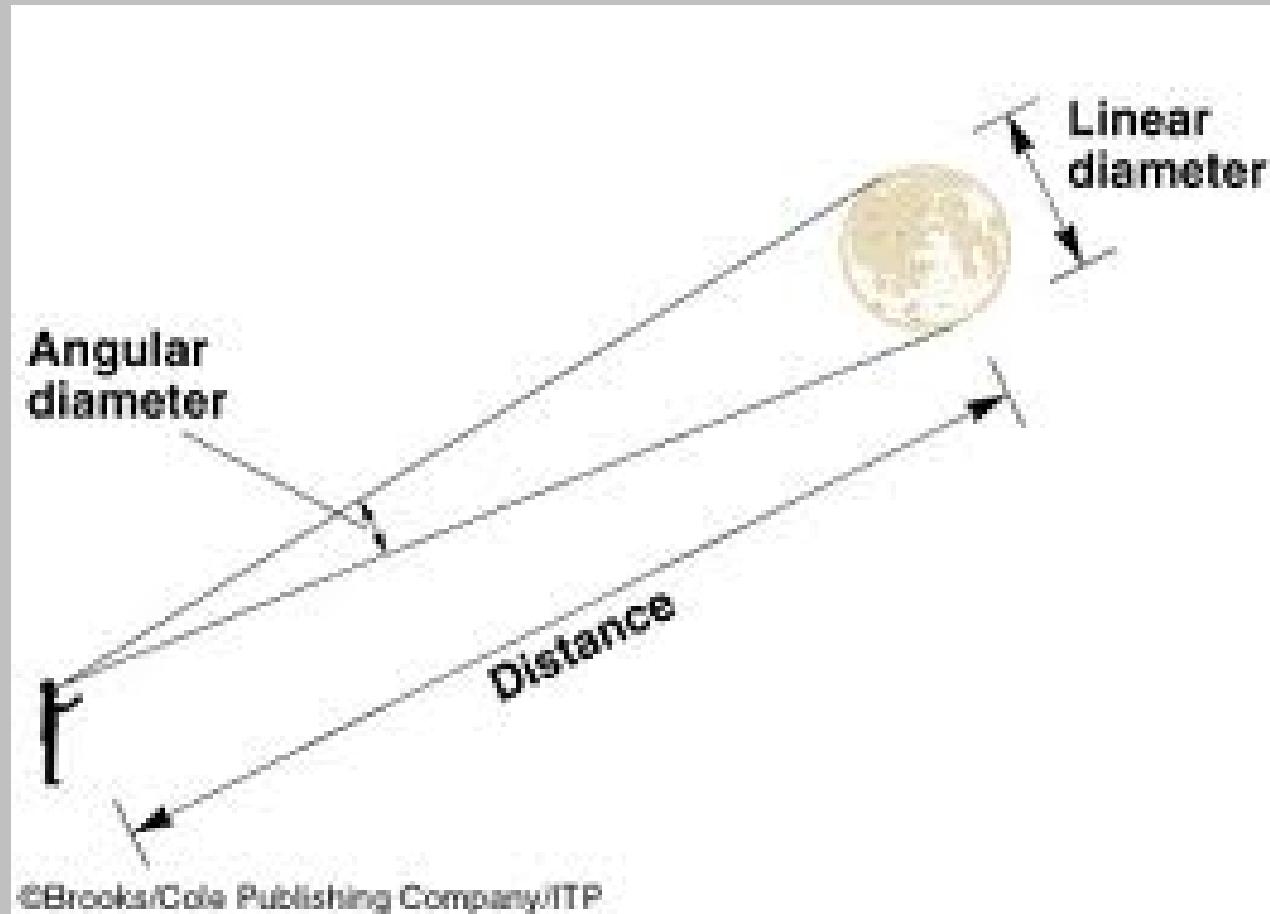
Angles, distances and widths

Calibrate using the Big Dipper!



Angles, distances and widths

Relationship between linear diameter and angular diameter



$$\text{AD(radians)} = \text{LD}/\text{D}$$

$$\text{AD(degrees)} = (57.3)\text{LD}/\text{D}$$